

Domino Startup Concept

Problem: Domino transplantation could save 2,000+ additional lives annually in the US, but only 50-100 domino transplants happen because:

Critical Data Corrections

Your PDF states that 50-100 domino transplants happen annually in the US. However, only **4 domino liver transplants** were performed in the entire United States in 2024 [NewYork-PresbyterianNewYork-Presbyterian](#). From 1996-2020, only 185 domino liver transplants occurred out of 181,976 total liver transplants (0.1%), averaging about 7-8 per year [ScienceDirectHPB](#).

SIMPLIFIED EXPLANATION FOR NON-BIOTECH AUDIENCE

The Problem (In Plain English)

Imagine you have a friend named Sarah who has a genetic condition called FAP (Familial Amyloid Polyneuropathy). Her liver works perfectly fine - it cleans her blood, produces proteins, does everything a liver should do. But there's one problem: it produces a slightly "wonky" protein that slowly builds up in her nerves over 20-30 years, eventually causing paralysis and death.

The only cure? Give Sarah a new liver. Her old liver gets removed.

Now here's the fascinating part: **Sarah's old liver is 99% perfect**. It would work great in someone else - like Michael, who's dying of liver failure from hepatitis and only has weeks to live.

This is called a "**domino transplant**" - one donor saves TWO lives:

- **Donor → Sarah** (Sarah gets cured of FAP)
- **Sarah → Michael** (Michael gets Sarah's functional liver)

Why Isn't This Happening More?

Out of 210 FAP patients who received transplants, only 103 (49%) had their livers reused. For another metabolic disease (MSUD), only 50 out of 227 livers (22%) were reused [ScienceDirectHPB](#).

The Current Manual Process (The Real Problem):

When a transplant coordinator identifies a potential domino donor like Sarah, they have to:

1. **Manually screen thousands of patients** on the waitlist to find someone like Michael
2. **Calculate complex risk-benefit tradeoffs** (Will Michael live long enough for the "wonky" protein to matter? Is he sick enough that this marginal liver is his best option?)
3. **Coordinate 3 simultaneous surgeries** across potentially different hospitals
4. **Get informed consent** from Michael and his family about accepting a "diseased" liver

This manual process takes 8-12 hours and is so complex that 30-40% of potential domino chains fall apart due to logistics [NewYork-Presbyterian](#).

CURRENT US HEALTHCARE SYSTEM WORKFLOW

Let me explain how organ transplants currently work:

The Standard System

UNOS (United Network for Organ Sharing) operates the national Organ Procurement and Transplantation Network (OPTN). When someone becomes an organ donor, their information goes into UNet, a centralized computer system that automatically generates a ranked list of compatible recipients based on blood type, urgency (MELD score), location, and waiting time [UNOS](#).

Transplant coordinators have just 1 hour to accept or decline an organ offer [Organ Donation and Transplantation Alliance](#).

Why Domino Transplants Don't Fit This System

Standard transplants are automated:

- Donor → Computer matches → Recipient
- Takes minutes

Domino transplants are manual:

- Donor A → Patient B (who becomes a secondary donor) → Patient C
- Requires human judgment: "Which of 10,000+ waitlist patients would benefit most from a liver with a 10-15 year 'expiration date' before disease manifests?"

Clinical transplant coordinators are nurses who arrange evaluations, coordinate care, and manage the entire transplant process. They're already overwhelmed with standard transplant coordination [OPTNUPMC](#).

THE REAL OPPORTUNITY & REFINED VALUE PROPOSITION

Updated Problem Statement

Problem: In 2024, only 4 domino liver transplants occurred in the US [NewYork-PresbyterianNewYork-Presbyterian](#), despite over 11,000 liver transplants being performed annually and approximately 500-800 metabolic disease patients receiving transplants who could serve as domino donors [OPTN](#). Historical data shows that only 49% of FAP livers and 22% of MSUD livers are reused as domino transplants [HPB](#).

Why? Transplant coordinators lack tools to:

1. Quickly identify which of 10,000+ waitlist patients are optimal domino candidates
2. Calculate personalized survival probabilities for patients accepting "marginal" livers
3. Coordinate complex 3-patient logistics within tight timeframes
4. Generate informed consent materials explaining the trade-off

Refined Solution

DominoMatch: An AI-powered decision support platform that turns domino transplantation from a

manual, 8-12 hour process into a streamlined, data-driven workflow, enabling transplant centers to identify optimal candidates in minutes, coordinate multi-patient logistics, and provide patients with personalized survival calculators for truly informed consent.

CRITICAL ISSUE YOUR PDF DOESN'T ADDRESS

There's a **major medical concern** with FAP domino transplants that you need to acknowledge:

Recent data from Portugal (which has the world's largest domino recipient population) shows that recipients are developing "acquired FAP" much faster than expected - within 3-10 years instead of the predicted 20-25 years [Neurology](#). In one Portuguese study, 23 out of

47 domino recipients became symptomatic, and pathological testing showed amyloid deposits in nearly all patients [Neurology](#).

What this means for your buildathon:

- You should focus on matching older patients (65+) with short life expectancies
- Your informed consent materials must clearly explain this accelerated timeline
- This is actually a STRENGTH of your product - helping identify patients who will die from their current condition before FAP manifests

HOW TO IMPLEMENT IN CURRENT US SYSTEM

Integration Points

1. **Data Source:** UNOS maintains UNet, which contains all waitlist patient data. Your platform would need to integrate with or receive data exports from UNet [UNOS](#)
2. **User:** Transplant coordinators at the 250+ transplant centers
3. **Workflow Integration:**

Current: FAP patient identified → Manual 8-12 hour screening → Often nothing happens

With DominoMatch: FAP patient identified → Upload to DominoMatch →

AI ranks top 10 candidates in minutes → Coordinator reviews →

Automated logistics coordination → AI-generated informed consent

4. **Regulatory Path:** This is a **clinical decision support tool**, not a medical device. It doesn't make decisions - it helps coordinators make better decisions faster.

Solution: AI-powered platform that identifies optimal domino recipients, calculates personalized survival probabilities, coordinates multi-patient logistics, and generates patient-friendly decision support materials.

Market Opportunity:

- 250+ transplant centers in US
- 30,000+ patients on liver transplant waitlist

- 500-800 FAP/metabolic disease patients transplanted annually (potential domino donors)
 - Current domino utilization rate: 10-15%
 - Target domino utilization rate: 40-50% = 1,500 additional lives saved/year
-

What You Build in 11 Hours

Core Features:

1. Domino Candidate Matcher (3 hours)

- Input: Upload Patient B details (metabolic disease type, liver function) + Patient C waitlist data (CSV)
- AI Analysis:
 - Scores each waitlist patient for domino suitability based on:
 - Age (60+ preferred)
 - MELD score (urgency)
 - Life expectancy without transplant (<6 months optimal)
 - Expected lifespan with domino liver (>10 years valuable)
 - Comorbidities that might kill them before metabolic disease manifests
 - Ranks top 10 candidates with explanations

2. Personalized Survival Calculator (3 hours)

- Interactive Tool:
 - Patient C enters their data (age, diagnosis, current health status)
 - Two scenarios displayed side-by-side:
 - Option A: Wait for standard donor (survival probability curve)
 - Option B: Accept domino liver (survival probability curve with metabolic disease onset marked)
 - Visual timeline showing: "You gain 12 years of life expectancy vs. waiting"

- Trade-off visualization: "90% chance you survive to see grandchildren graduate vs. 30% chance waiting for standard donor"

3. Informed Consent Generator (2 hours)

- AI-Powered Document Creation:
 - Generates patient-friendly consent forms explaining:
 - What domino transplant is (layman's terms with diagrams)
 - Specific risks for their metabolic disease type
 - Personalized timeline of when symptoms might appear
 - Comparison to "do nothing" scenario
 - Available in multiple languages
 - Includes visual infographics

4. Domino Chain Coordinator Dashboard (2 hours)

- Timeline Management:
 - Tracks all three parties: Donor A → Patient B → Patient C
 - Synchronizes surgical schedules across teams
 - Alerts for timing conflicts
 - Organ transport logistics
 - Color-coded status tracker (Green: on schedule, Red: delay risk)

5. Bonus Feature: Outcome Predictor (1 hour if time permits)

- Upload historical domino transplant data
- ML model predicting success probability based on patient characteristics
- "Patients like you have 85% 10-year survival rate"

11-Hour Build Plan

Hour 0-1: Setup & Data

- Set up Next.js + Tailwind project

- Find/create sample datasets:
 - Mock waitlist data (50 patients with MELD scores, ages, diagnoses)
 - FAP disease progression data (from medical literature)
 - Survival curve data for liver failure patients
- Set up OpenAI API access

Hour 1-3: Matching Algorithm

Frontend:

- Upload interface for Patient B profile + waitlist CSV
- Results table showing top 10 matches

Backend:

- Scoring algorithm considering:
 - Age weight (60-75 = high score)
 - MELD score (higher = more urgent = higher score)
 - Comorbidity risk (diabetes, heart disease = higher score - likely to die before metabolic disease)
 - Blood type compatibility
- GPT-4 generating natural language explanations: "John Smith (68) is optimal because..."

Hour 3-6: Survival Calculator

Frontend:

- Clean, interactive interface with dual survival curves
- Slider inputs for patient parameters
- Real-time graph updates
- Export to PDF button

Backend/Logic:

- Mathematical models for survival probability:
 - Standard waitlist survival (exponential decay based on MELD)

- Domino liver survival (plateau for 10-15 years, then decline)
- Chart.js for beautiful visualizations
- Calculate key metrics:
 - "Expected additional years of life"
 - "Probability of surviving to age X"
 - "Risk of metabolic symptoms by year"

Hour 6-8: Consent Generator + Dashboard

Consent Generator:

- Template-based generation with patient-specific data insertion
- GPT-4 creating patient-appropriate explanations
- Visual timeline generator
- PDF export with diagrams

Dashboard:

- Kanban-style board showing domino chain progress
- Three columns: Patient A (donor), Patient B (recipient/donor), Patient C (final recipient)
- Timeline view with surgical scheduling
- Alert system for conflicts

Hour 8-9: Demo Preparation

- Create 3 realistic patient scenarios
- Record demo video walkthrough
- Screenshot key features
- Prepare sample outputs (PDFs, reports)

Hour 9-10: Pitch Deck

- Problem slide (with statistics on wasted domino opportunities)
- Solution demo (screenshots + live demo plan)

- Market size
- Business model
- Traction plan (pilot with 3 transplant centers)

Hour 10-11: Polish & Practice

- Fix bugs
- Improve UI/UX
- Practice 5-minute pitch
- Prepare for Q&A

Tech Stack (All Feasible in 11 Hours)

Frontend:

- Next.js + React
- Tailwind CSS for rapid UI
- Chart.js or Recharts for survival curves
- React PDF for document generation

Backend/AI:

- OpenAI GPT-4 API for matching explanations and consent generation
- Simple scoring algorithms (no complex ML training needed)
- Firebase or Supabase for quick database (optional)

Data:

- Mock datasets (CSV files)
- Public medical literature for disease progression rates
- UNOS (organ network) publicly available statistics

Demo Strategy for Judges

Opening Hook (30 seconds):

"Every year, 400 people donate organs that could save TWO lives instead of one. But we only capture 50 of these opportunities because hospitals can't efficiently match domino recipients. DominoMatch uses AI to turn one donor into two survivors."

Live Demo (3 minutes):

Scenario 1 - The Matching:

1. "This is Sarah, 42, with FAP. She's getting a new liver today."
2. Upload Sarah's profile + waitlist of 50 patients
3. AI instantly identifies top 10 candidates
4. Click on Michael, 67, with acute liver failure
5. Show explanation: "Michael is optimal because he's 67 (limited life expectancy), MELD score 32 (will die in 2 weeks without transplant), and has diabetes (likely to die before FAP symptoms emerge in 15 years)"

Scenario 2 - The Decision Support:

1. Michael's family wants to understand the trade-off
2. Open Survival Calculator
3. Show two curves side-by-side:
 - Red curve: "Wait for standard donor" → 15% chance of surviving 6 months
 - Green curve: "Accept domino liver" → 85% chance of surviving 10+ years
4. Highlight: "Michael gains 9.4 expected years of life"
5. Generate informed consent PDF in real-time

Scenario 3 - The Coordination:

1. Show dashboard tracking all three patients
2. Demonstrate scheduling conflict detection
3. "Surgery must happen within 8-hour cold ischemia time—our platform ensures perfect coordination"

Closing Impact Statement:

"DominoMatch could increase domino transplant utilization from 10% to 50%, saving an additional 1,500 lives annually in the US alone. At \$200/month per transplant center, that's a \$600K ARR market with lives saved as our North Star metric."

Business Model

Revenue Streams:

SaaS Model:

- \$200/month per transplant center (250 centers in US = \$600K ARR potential)
- Enterprise tier: \$500/month (includes API access, custom integrations)

Per-Transaction Model (Alternative):

- \$500 per successful domino match facilitated
- Aligns incentives with outcomes

Data Licensing:

- Anonymized domino transplant outcome data to researchers
- \$50K-100K annually

Go-to-Market:

Phase 1 (Months 1-6): Pilot with 3 high-volume transplant centers

- Focus on FAP domino liver transplants (easiest use case)
- Prove 3x increase in domino utilization

Phase 2 (Months 6-12): Expand to 20 centers

- Add heart-lung domino capabilities
- Build integrations with UNOS (national organ network)

Phase 3 (Year 2): National rollout

- Partner with transplant societies
 - Publish clinical outcomes data
 - Expand internationally (Europe has high domino adoption)
-

Competitive Advantages

1. First-mover in domino-specific software - No existing specialized tools
2. AI-powered matching - Faster and more objective than manual review
3. Patient-facing decision support - Improves informed consent process
4. Lives-saved metric - Measurable impact attracts funding/partnerships
5. Network effects - More centers = better data = better matching algorithms